AP Calculus BC Unit 4 Review Book

Contextual Applications of Differentiation

This review book is written as a beginner-friendly but fully detailed final review for AP Calculus BC Unit 4. It covers every major topic in the official unit: meaning of the derivative in context, motion along a line, rates of change in nonmotion settings, related rates, local linearity and linearization, and L'Hospital's Rule. The emphasis is on what each quantity means, how to set up each type of problem, and how to avoid the mistakes that cost points on the AP exam.

Unit 4 topic	Core AP skill
4.1	Interpret a derivative as an instantaneous rate of change using a full sentence and correct units.
4.2	Connect position, velocity, acceleration, speed, displacement, and total distance.
4.3	Transfer derivative meaning to population, volume, temperature, revenue, and other real contexts.
4.4-4.5	Set up and solve related-rates problems from geometry and measurement situations.
4.6	Use linearization to estimate nearby values and understand why the estimate works.
4.7	Recognize valid indeterminate forms and use L'Hospital's Rule correctly.

First memory checkpoint. Before you start mixed practice, you should already know the motion formulas, the meaning of speed, the related-rates workflow, the linearization formula, and the only indeterminate forms that allow direct use of L'Hospital's Rule.

$$v(t) = s'(t)$$

$$a(t) = v'(t) = s''(t)$$

$$\text{speed} = |v(t)|$$

$$L(x) = f(a) + f'(a)(x - a)$$

Valid direct use of L'Hospital: $\frac{0}{0}$ or $\frac{\infty}{\infty}$

4.1 Interpreting the Meaning of the Derivative in Context

A derivative in context is an **instantaneous rate of change**. In Unit 4, this means you must move beyond 'the derivative is the slope' and explain exactly what is changing, with respect to what, at which input value, and with what units.

A fully correct AP-style interpretation always includes four ingredients: the input value, the quantity changing, the direction of change, and the units. If any of these are missing, the explanation is incomplete.

General interpretation template

If *Q(t)* is a quantity and *Q'(a)=k*, then: at time *t=a*, the quantity *Q* is increasing or decreasing at *k* units of output per unit of input.

The sign of the derivative tells the direction of change: positive means increasing; negative means decreasing.

The units of a derivative are always 'units of output per unit of input'.

Example

Suppose *P(t)* is population measured in thousands of people and *P'(4)=-2.5*. A correct interpretation is: *At time 4, the population is decreasing at 2.5 thousand people per year* (or whatever the time unit is).

Average rate versus instantaneous rate

$$\text{Average rate of change on } [a, b] = \frac{f(b) - f(a)}{b - a}$$

$$\text{Instantaneous rate of change at } x = a = f'(a)$$

Students often confuse these. Unit 4 context questions almost always want the second idea, not the first.

4.2 Straight-Line Motion: Position, Velocity, and Acceleration

If a particle moves along a line and its position is described by *s(t)*, then derivatives and integrals connect every motion quantity.

$$v(t) = s'(t)$$

$$a(t) = v'(t) = s''(t)$$

$$s(b) - s(a) = \int_a^b v(t)\,dt$$

$$v(b) - v(a) = \int_a^b a(t)\,dt$$

What each quantity means

Position: where the particle is located on the line.

Velocity: how position is changing, including direction.

Acceleration: how velocity is changing.

Speed: the magnitude of velocity.

$$\text{speed} = |v(t)|$$

Direction and motion language

If $v(t) > 0$, the particle moves to the right.

If $v(t) < 0$, the particle moves to the left.

If $v(t) = 0$, the particle is momentarily at rest.

Speeding up and slowing down

If velocity and acceleration have the same sign, the particle is speeding up.

If velocity and acceleration have opposite signs, the particle is slowing down.

Worked example

Let

$$s(t) = t^3 - 6t^2 + 9t$$

Then

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

At $t=1$,

$$v(1) = 0, \qquad a(1) = -6$$

So the particle is momentarily at rest at $t=1$. Because the acceleration is negative, the velocity is decreasing at that instant.

Displacement versus total distance

$$\text{displacement on } [a, b] = \int_a^b v(t)\,dt$$

$$\text{total distance on } [a, b] = \int_a^b |v(t)|\,dt$$

If velocity changes sign, split the interval at each time where *v(t)=0* before evaluating total distance. This distinction is one of the most frequently tested motion ideas.

4.3 Rates of Change in Other Applied Contexts

The derivative is not only about motion. The same idea appears in population models, production, water flow, temperature change, profit, cost, and growth or decay settings. The mathematics is identical even when the story is different.

If *R(t)* is revenue in dollars, then *R'(t)* is revenue change in dollars per unit time.

If *V(h)* is volume in cubic centimeters as a function of height, then *V'(h)* is cubic centimeters per centimeter.

If *C(x)* is cost in dollars for producing *x* items, then *C'(x)* is the marginal cost near that production level.

The main AP skill is interpretation. Your sentence should name both variables and include units.

4.4 Introduction to Related Rates

A related-rates problem has several variables changing at the same time, usually through a geometry formula. You write the relationship first, then differentiate with respect to time.

The five-step related-rates method

1. Draw a diagram and define every variable.
2. Write the equation that relates the variables.
3. Differentiate both sides with respect to time *t*.
4. Substitute the numerical values for the requested instant.
5. Solve for the unknown rate and state units.

Never substitute numerical values before differentiating unless a quantity is truly constant. This is the most common setup error.

Common geometry formulas

$$A = \pi r^2$$

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3$$

$$V_{\text{cylinder}} = \pi r^2 h$$

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

$$x^2 + y^2 = z^2$$

4.5 Solving Related Rates Problems

Example 1: expanding circle

Suppose the radius of a circle increases at 3 cm/s. How fast is the area changing when the radius is 5 cm?

$$A = \pi r^2$$

Differentiate with respect to time:

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Substitute *r=5* and *rac{dr}{dt}=3*.

$$\frac{dA}{dt} = 2\pi(5)(3) = 30\pi$$

So the area is increasing at **30π square centimeters per second**.

Example 2: ladder problem

A ladder of fixed length 10 ft leans against a wall. The bottom slides away from the wall at 2 ft/s. Find how fast the top slides down when the bottom is 6 ft from the wall.

$$x^2 + y^2 = 10^2$$

Differentiate:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$x\frac{dx}{dt} + y\frac{dy}{dt} = 0$$

When *x=6*, solve for *y* first:

$$6^2 + y^2 = 100 \Rightarrow y = 8$$

Now substitute:

$$6(2) + 8\frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\frac{3}{2}$$

The top is sliding downward at **1.5 ft/s**.

Example 3: sphere volume

A spherical balloon is inflated so that the radius increases at 0.5 cm/s. Find the rate of change of the volume when the radius is 4 cm.

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$$

$$\frac{dV}{dt} = 4\pi(4)^2(0.5) = 32\pi$$

The volume is increasing at **32π cubic centimeters per second**.

Related-rates checklist

Differentiate before plugging in values.

If the problem asks for a rate at a certain instant, you may need to find another variable value first.

Pay attention to negative signs: 'downward' or 'decreasing' often corresponds to a negative derivative.

Always include units.

4.6 Local Linearity and Linearization

A differentiable function looks almost like a line when you zoom in near a point. The tangent line becomes a practical approximation to the function near that point. This is called **linearization**.

$$L(x) = f(a) + f'(a)(x - a)$$

Near *x=a*, the approximation is

$$f(x) \approx L(x)$$

Worked example: estimate a square root

Estimate √*4.1* using the function *f(x)=\sqrt{x}* at *a=4*.

$$f(x) = \sqrt{x}, \quad f(4) = 2, \quad f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(4) = \frac{1}{4}$$

$$L(x) = 2 + \frac{1}{4}(x - 4)$$

$$L(4.1) = 2 + \frac{1}{4}(0.1) = 2.025$$

Therefore

$$\sqrt{4.1} \approx 2.025$$

Differentials viewpoint

$$dy = f'(x)\,dx$$

$$\Delta y \approx dy = f'(a)\Delta x$$

This is the same local-linearity idea written in differential form.

When the estimate is an overestimate or underestimate

If the graph is concave up, the tangent line usually lies below the graph near the center point, so the linearization is usually an underestimate.

If the graph is concave down, the tangent line usually lies above the graph near the center point, so the linearization is usually an overestimate.

4.7 L'Hospital's Rule

L'Hospital's Rule is used to evaluate certain difficult limits, but it is valid only for quotients that produce an indeterminate form of **0/0** or **infinity/infinity**.

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)} \quad \text{if the original form is } \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

Valid direct forms

\(\frac{0}{0}\)

\(\frac{\infty}{\infty}\)

Not valid directly

\(0\cdot\infty\)

\(\infty-\infty\)

\(1^\infty,\ 0^0,\ \infty^0\)

These must be rewritten first.

Example 1

$$\lim_{x\to 0}\frac{\sin x}{x}$$

Direct substitution gives *0/0*, so L'Hospital applies.

$$\lim_{x\to 0}\frac{\cos x}{1}=1$$

Example 2

$$\lim_{x\to\infty}\frac{\ln x}{x}$$

Direct substitution gives *infinity/infinity*.

$$\lim_{x\to\infty}\frac{1/x}{1}=0$$

Example 3: rewrite first

$$\lim_{x\to\infty}xe^{-x}$$

Rewrite as a quotient:

$$\lim_{x\to\infty}\frac{x}{e^x}$$

Now the form is *infinity/infinity*.

$$\lim_{x\to\infty}\frac{1}{e^x}=0$$

L'Hospital warning list

Always identify the indeterminate form first.

If the derivative quotient still has an indeterminate form, you may apply the rule again.

If simple algebra or factorization is easier, use that instead.

Do not use L'Hospital on a quotient that is not indeterminate.

Unit 4 Final Summary

The entire unit can be reduced to a few stable ideas. A derivative in context is a rate. Motion formulas connect position, velocity, and acceleration. Related-rates problems require a geometry equation before differentiation. Linearization uses the tangent line as a local approximation. L'Hospital's Rule applies only to the right kinds of indeterminate quotients.

Final formula set to memorize

$$v(t) = s'(t)$$

$$a(t) = v'(t) = s''(t)$$

$$\text{speed} = |v(t)|$$

$$\text{displacement} = \int_a^b v(t)\,dt$$

$$\text{distance} = \int_a^b |v(t)|\,dt$$

$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$$

$$\frac{dV_{\text{sphere}}}{dt} = 4\pi r^2\frac{dr}{dt}$$

$$L(x) = f(a) + f'(a)(x-a)$$

Most common AP mistakes

Confusing average rate of change with instantaneous rate of change.

Leaving units out of a context interpretation.

Using velocity when the question asks for speed, or vice versa.

Deciding that an object is speeding up just because acceleration is positive.

Substituting numerical values before differentiating in a related-rates problem.

Applying L'Hospital's Rule without first checking that the form is valid.

End of Unit 4 review book.